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UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING
27 JUL 2025
EXAMINATIONS & REGISTRATION

UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

Master of Information Technology

First Semester Examination – 2025

MIT 1304 - Data Communications and Computer Networks

Two (2) Hours

July/Aug. 2025

To be completed by the candidate								
Index No:								

Important Instructions to candidates

- Please ensure that you have the correct examination paper in front of you.
- Students should answer in the medium of **English language only** using the space provided in this question paper.
- Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- Clearly write your index number on each page of this Question paper.
- This paper consists of **04** questions in **12** pages (including the Cover Page).
- The duration of the paper is Two (2) Hours.
- Answer **all** questions.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the Examination Hall by a candidate.
- Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.
- **Non-Programmable** calculators are **allowed**.

For Examiner's use only	
Question No.	**Marks**
1	
2	
3	
4	
Total	

Index No.								

Q1. A. A signal is produced by varying the intensity of a specific type of energy over time through an appropriate medium. Write down three (3) types of energy that can be used for this purpose.

(03 marks)

ANSWER BOX

Q1. B. What does the term '**throughput**' refer to in the context of a communication channel, and how does it differ from the channel's **bandwidth**?

(05 marks)

ANSWER BOX

Q1. C. What is the purpose of Pulse Code Modulation (**PCM**) in digital communication systems?

(02 marks)

ANSWER BOX

Index No.								

Q1. D. A communication link has a data rate of 10 Mbps and a propagation delay of 50 milliseconds.

(a) Calculate the **delay-throughput product** for this link.

(b) Explain what the delay-throughput product represents in this context.

(05 marks)

ANSWER BOX

Q1. E. (a) What is **Manchester Encoding** in digital communication?

(b) Using Manchester Encoding, draw the timing diagram for the 8-bit binary pattern: **10101100**. Assume the signal starts with a high-to-low transition for a '**0**' and a low-to-high transition for a '**1**'.

(05 marks)

ANSWER BOX

Index No.								

Q1. F. (a) How does a Layer 2 switch build its **Forwarding Table** in a Local Area Network (LAN)? Explain the process briefly.

(b) What are Layer 2 loops in a LAN, and why are they problematic?
If such loops exist in the network, what mechanisms are used to detect and prevent them?

(05 marks)

ANSWER BOX

Q2. A. Explain the difference between **Physical** and **Logical** LAN topologies. Provide **one** example for each to illustrate your answer.

(05 marks)

ANSWER BOX

Index No.								

Q2. B.

(a) In the Ethernet (or IEEE 802.3) frame format, the **Destination MAC address** appears as the **first** field in the frame header. Why is it important to place the destination address at the beginning of the frame?

(b) Draw and label the Ethernet frame header format, clearly indicating the size of each field in bytes.

(05 marks)

ANSWER BOX

Q2. C.

When one host in a Local Area Network (LAN) wants to communicate with another using IP, it needs the destination's MAC address for the Ethernet frame.

(a) Describe what happens when the destination MAC address is not already known to the sender.

(b) What kind of message is sent and how is it delivered on the LAN?

(05 marks)

ANSWER BOX

Index No.								

Q2. D. Describe the type of delivery service provided by the Internet Protocol (IP). Additionally, list three (3) key functions performed by the IP layer in the TCP/IP model.

(05 marks)

ANSWER BOX

Q2. E. Suppose a host sends an IP datagram to a remote destination, and the datagram must pass through a router (R_{10}) that encounters a routing issue where the next hop is unreachable. Explain how this situation involves the use of the **ICMP error** reporting protocol and what message is sent back to the sender.

(05 marks)

ANSWER BOX

Index No.								

Q3. A. Answer the following questions about the IPv4 header.

(a) The value of the **Header Length** field in an IPv4 packet is **1101**. Calculate the total length (in bytes) of the IPv4 header and determine the size (in bytes) of the **Options** field.

(b) Describe how a router can be configured to block all **UDP** traffic entering a specific network using IPv4 filtering mechanisms.

(06 marks)

ANSWER BOX

Q3. B. What is the **aggregated** IP block (Supernet) that encompasses the consecutive IP blocks from **144.0.0.0/16** to **147.255.0.0/16**?

Justify your answer by showing the necessary binary calculations and explain the number of IP blocks aggregated.

(04 marks)

ANSWER BOX

Index No.								

Q3. C. An IP datagram with a payload of **3280 bytes** must be transmitted over a link with a Maximum Transmission Unit (**MTU**) of **1024** bytes. Assume that the IP datagram does not use any Options, and the Identification number is **0x5BC2**. The Don't Fragment (DF) bit is **not set**.

After fragmentation, complete the following table with the values for each resulting fragment. Corresponding fragments are numbered as 1, 2, 3, ... in the table to denote each fragment.

(06 marks)

ANSWER BOX

Fragment No.	Total Length	Fragment Offset	MF Flag

Q3. D. Your organization has deployed a **private** IP network using the address range 192.168.1.0/24. The internal users need to access the Internet, but the network has only one **public** IP address available.

(a) Explain how Network Address Translation (NAT) enables multiple internal hosts to access the Internet using the single public IP address.

(b) Identify one major limitation of traditional NAT and how it impacts certain applications.

(04 marks)

ANSWER BOX

Index No.								

Q3. E. Answer the following questions about the IPv6 header.

(a) Write down the IPv6 address 2000:0000:020A:1A2B:1A2B:0000:0000:0000 in abbreviated form.

(b) Write down the 64-bit IEEE Extended Unique Identifier (EUI) for the MAC Address 41-21-44-66-12-34.

(05 marks)

ANSWER BOX

Q4. A. Differentiate between the Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) in terms of their key characteristics and use cases.

(05 marks)

ANSWER BOX

Index No.								

Q4. B. You have been hired to design the IP addressing scheme for a company that has a Head Office and **two** Branch Offices. All locations are interconnected using leased lines, and the company has been allocated the IP block **192.168.16.0/21** for its internal network.

The number of hosts required at each site is as follows:

Head Office: **400** hosts
Branch Office A: **200** hosts
Branch Office B: **100** hosts

In addition to assigning IP subnets for each LAN, you are required to allocate IP addresses for the leased lines connecting: Head Office and Branch Office A, Head Office and Branch Office B as well.

(a) How many different hosts in the 192.168.16.0/21 address range have machine assignable IP addresses accessible to them?

(b) Write down for each Subnet, its network address, broadcast address, and the correct subnet mask in CIDR form in the table provided in the answer box.

(c) Write down the unallocated IP block available after the above allocation, with its network address, broadcast address and the correct subnet mask in CIDR form in the table provided in the answer box.

(10 marks)

ANSWER BOX

(a) Machine assignable IP addresses in 192.168.16.0/21 =

(b) Fill the table given below.

Subnet No	Network Address	Broadcast Address	Subnet mask (in CIRD)
Head Office			
Branch Office A			
Branch Office B			
Head Office to Branch Office A			
Head Office to Branch Office B			

Continued on next page ...

Index No.								

(c) Fill the table given below. Add more rows if required.

Block No	Network Address	Broadcast Address	Subnet mask (in CIRD)

Index No.								

Q4. C. Explain the mechanisms **TCP** uses to provide **reliable** data transmission. Additionally, describe how TCP supports **end-to-end** communication between applications.

(05 marks)

ANSWER BOX

Q4. D. Describe the process by which a web browser initiates an **HTTP** session after a user enters a URL into the address bar. Additionally, define what is meant by a **persistent HTTP connection** and explain how it enhances web performance.

(05 marks)

ANSWER BOX

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